

Muhammad Shahroz Butt

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EDUCATION

University of Central Punjab

Nov. 2022 – Aug 2026

B.S. Computer Science | GPA: 3.33/4.0

Lahore

- Courses: Computer Communication and Networking, Artificial Intelligence, Operating Systems, Data Structures, Programming Fundamentals, Object Oriented Programming, Algorithms & Analysis, Data Science, Information Security

TECHNICAL SKILLS

Languages: C/C++, JavaScript, TypeScript, Python, SQL, HTML, CSS

Technologies: React Native, React.js, Vite, Django, Django REST Framework, JWT, WebSockets (Django Channels), MongoDB, Redis, PyTorch, Hugging Face Transformers, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, NLP

Tools: VS Code, Git/GitHub, Postman, Jupyter Notebook, ESLint

EXPERIENCE

Data Science & Analytics Intern | DevelopersHub Corporation

Mar 2026 – Apr 2026

- Successfully completed a 6-week internship program in Data Science & Analytics with exceptional performance.
- Contributed through practical analytics and machine learning work during the internship period (March 1st, 2026 to April 12th, 2026).

RESEARCH

Fake News Detection Research | Python, PyTorch, Hugging Face Transformers, scikit-learn, Pandas, NumPy

- Aggregated and processed public fake-news datasets, then ran reproducible stratified 10k and 30k experiments to benchmark transformer models for binary classification.
- Evaluated RoBERTa-base, XLM-RoBERTa-base, RemBERT, TinyBERT, MobileBERT, SciBERT, DistilBERT, ConvBERT, AdaBERT, and FastBERT across accuracy, precision, recall, F1, AUC, and MCC.
- Documented the strongest 10k result with RoBERTa-base (89.13% accuracy, 89.78 F1) and the strongest 30k balance with XLM-RoBERTa-base (88.62% accuracy, 89.59 F1).

PROJECTS

Iris Dataset EDA | Python, Pandas, Matplotlib, Seaborn

- Performed exploratory analysis on the Iris dataset with scatter plots, histograms (KDE), and species-wise box plots to identify class patterns and feature separability.

Credit Risk Prediction | Python, Scikit-learn, Pandas, Seaborn

- Built a classification pipeline on loan default data using Logistic Regression and Decision Tree; achieved 0.89 accuracy with Logistic Regression and compared results using confusion matrices.

Customer Churn Prediction | Python, Scikit-learn, Pandas

- Developed churn prediction models with preprocessing (encoding + scaling) and model comparison; Random Forest achieved 0.866 accuracy and identified top churn drivers such as Age, Balance, and CreditScore.

Bank Marketing Explainable ML | Python, Scikit-learn, SHAP

- Predicted term-deposit subscription using Logistic Regression and Random Forest, then explained model outputs with SHAP force plots; Random Forest achieved stronger F1 and ROC-AUC performance.

TRAK | React Native, React.js, Vite, Django, Django REST Framework, JWT, MongoDB, Redis, PyTorch
[<https://trak-fyp.netlify.app/>] (Ongoing)

- Developed a cross-platform news aggregation platform for Android/iOS/Web with React Native (mobile) and React + Vite (web), including authentication, personalized feeds, search, and an admin analytics dashboard.
- Built backend APIs with Django REST Framework and JWT-based auth, configured CORS and MongoDB integration, and implemented real-time updates using Django Channels with Redis.
- Integrated automated news ingestion using feed parsing and web scraping, and extended the pipeline with ML components for content credibility scoring using PyTorch and Transformers.

Pine-Tree House | HTML5, CSS, Vanilla JavaScript, Node.js, MySQL [<https://pine-tree-house.netlify.app/>] (Ongoing)

- A full-stack booking platform for a premium villa rental business featuring a responsive website with interactive galleries and real-time availability. Built a Node.js/Express backend with MySQL database, Stripe payment processing, and automated notifications via email and SMS. Implemented admin dashboard for managing reservations, pricing, and guest information with security features including JWT authentication, password encryption, and rate limiting. The system handles the complete booking lifecycle from inquiry to payment confirmation with seamless frontend-backend synchronization.

Msquizflix | React.js, CSS, JavaScript [https://msquizflix.netlify.app/]

- Msquizflix responsive quiz web application, which simulates a quiz environment where user can answer multiple-choice questions divided into different sections, similar to a university entrance test.

Tailorized | React.js, Node.js, Express.js, SQLite3, CORS, Electron.js [https://tailorized-client.vercel.app/]

- A desktop application built with Electron, React, and Node.js that allows tailor shops to manage customers, measurements, and orders efficiently.

Milkish | CSS, HTML, Dynamic Data Store, JavaScript [https://mbudget.netlify.app/]

- A user-friendly platform designed to help users efficiently track and manage their monthly milk expenses. This application leverages modern web technologies, including Vue.js for the frontend, Express.js for the backend, and MongoDB as the database, to provide a seamless and interactive user experience.

STEM- Result Generator | Vue.js, TypeScript, Tailwind CSS

- A Vue.js-based form for student marks entry with features to dynamically generate a PDF report. The form allows users to input student details such as name, month, class, number of subjects, and corresponding marks. A table dynamically populates for marks entry based on the number of subjects, enabling seamless user interaction. The PDF generation leverages jsPDF and jspdf-autotable libraries to create a professional report that includes: Student details, Subject-wise marks breakdown, Total marks, Obtained marks, Percentage, A random remark based on the percentage, A section for a signature. The user interface is designed with Tailwind CSS for responsiveness and aesthetic appeal, ensuring usability and accessibility. This project combines modern front-end frameworks and libraries to streamline academic data management, offering scalability and customization for diverse educational applications.

Retail Bill System | Python, Tkinter

- A Billing Management System designed using Python's Tkinter library. It is a desktop-based application tailored for professional use, allowing businesses to manage customer details, generate bills, and calculate product prices with taxes. The system includes modules for tracking cosmetics, grocery items, and beverages. Features include: Bill Generation: Automatically generates a bill with itemized details, tax calculations, and a total amount. Search Functionality: Allows users to retrieve past bills using a unique bill number. Save and Print: Bills can be saved in a structured format for future reference. User-friendly Interface: Built with Tkinter's LabelFrames and Buttons for an intuitive experience. This application is ideal for small to medium-sized businesses looking for a simple and efficient billing solution.

Personal Portfolio | Next.js, TypeScript, React, Tailwind CSS, PostCSS, shadcn/ui [https://stb-dev.netlify.app/]

- A modern Next.js 15 portfolio built with TypeScript, Tailwind CSS 4, and shadcn/ui components powered by Radix UI primitives. It features Framer Motion animations, React Hook Form with Zod validation, theme switching, and analytics integration.